

BrightLearn Data Analytics

Research Assignment 1.

Section A: Data Fundamentals

1) What is a Database?

→ A database is an organized collection of information stored ~~elect~~ electronically so it can be easily accessed, managed & updated.

→ Databases are important because businesses deal with large amounts of information everyday. They make information easier to find & keep organised.

→ Examples - Bank database that stores customer accounts & transactions & Hospital database that stores patient information & medical records

2) What is Data Warehouse?

→ A central place where data from different sources is collected, cleaned & stored mainly for reporting & analysis.

Database

→ Usually used for everyday transactions such as recording payment & analysis.

→ A company would use one when it needs information from different systems in one place to support better decisions.

Datawarehouse

→ Mainly used to analyse historical information & find trends

3) What is a Data Lake

→ A large storage area that can hold data in its original or raw form.

→ It can store structured, semi-structured & unstructured data.

Datawarehouse

→ Normally contains cleaned & organised data

Data Lake

→ Can keep raw data before it's processed

→ A business may prefer a data lake when it has very large amounts of different types of data & wants flexibility for analytics, AI or machine learning.

(4) What is Data Mart?

- A smaller section of a data warehouse that focuses on one department or business area.
- A company could have a finance data mart containing expenses, invoices, payments.
- Finance employees & managers would typically use it because it gives them the information they need without searching through the whole warehouse.

(5) Structured Data.

- Organised into a fixed format, usually rows & columns.
- e.g. Spreadsheet containing employees names, salaries, & departments.

b) Semi-Structured Data.

- Does not follow a strict table format but still has organisation tags, such as JSON or XML.

c) Unstructured Data.

- Does not have a fixed structure, for example, photographs, videos, audio recordings & free-form documents.

(6) What is data Modelling?

- Is the process of planning how data will be organised, stored & related to other data in a system.
- It is important because a good model makes information easier to store, find & analyse & can reduce duplicated or inconsistent information.
- Two types are the relational model, which stores information in tables & the dimensional model, which uses fact & dimension tables & is common in analytics.

(7) What is Data Mining?

- Is the process of examining large amounts of data to discover useful patterns, relationships & trends.
- Businesses use it to understand customers, identify risks & improve decisions.
- e.g., a supermarket can analyse purchases to find products often bought together.
- Bank can analyse transaction patterns to identify unusual activity that could indicate fraud.

(8) What is DBMS

- Is a software that allows users & applications to create, store, manage & retrieve information from database.

• MySQL

- A popular relational system often used for websites & business applications.

• PostgreSQL

- Is an open-source relational database suitable for complex applications & analytics.

• Oracle

- Widely used by large organisations that need strong performance & security.

• MongoDB

- Is a NoSQL database that stores flexible documents & is useful when data structures can change frequently.

(9) What is ETL

- a) **E** → Extract means collecting data from sources such as databases, spreadsheets or applications.
- b) **T** → Transform means cleaning & changing the data into useful & consistent format, such as removing duplicates or correcting values.
- c) **L** → Load means putting the transformed data into destination such as data warehouse.

→ ETL is important because analysts need clean & ~~reliable~~ reliable data before producing meaningful reports & insight.

(10) Common Data Storage formats:

(a) **CSV** (Comma-separated values) - Stores data in simple table-like format & is useful for spreadsheets & simple datasets.

(b) **JSON** (JavaScript Object Notation) - Stores data using objects & key-value pairs & is commonly used with APIs.

(c) **Parquet** - a column-based ~~data~~ format designed for efficient storage & analytics, especially with large datasets.

(d) **Avro** - a data serialisation format useful when data is exchanged between systems.

* **CSV** used for simple data, **JSON** for APIs, & **Parquet** or **Avro** for larger data-processing systems.

Section B

(11) What is Artificial Intelligence (AI)

→ ~~Computer or machines~~

→ is the technology that allows computers or machines to perform tasks that normally require human intelligence, such as understanding language, recognizing images & making decisions.

* **Narrow AI**

→ is designed for specific tasks.

→ e.g. Recommendation system that suggests movies

* **General AI**

→ General AI would be able to perform a wide range of intellectual tasks similar to human.

(12) What is Machine Learning (ML)

→ is a part of ~~AI~~ AI where computers learn patterns from data & use those patterns to make predictions or decisions.

→ Traditional programming normally requires rules & instructions from a programmer, while machine learning learns patterns from data.

→ a) Supervised learning

→ uses labelled examples, such as predicting spam.

(b) Unsupervised learning

→ finds patterns or groups without known answers, such as customer grouping.

(c) Reinforcement learning

→ Learns through rewards & penalties, such as in games or robotics.

(13) What is Deep learning.

→ is a type of Machine learning that uses artificial neural network with many layers to learn complex patterns.

* The relationship can be shown as $AI \rightarrow ML \rightarrow DL$.

→ Neural networks contain connected nodes that process information & learn patterns from data.

→ Deep learning is useful for image recognition, speech recognition & language processing.

(14) What is Natural Language Processing (NLP)

→ is a part of AI that allows computers to work with human language.

→ E.g. - Voice assistants

- Translation applications

- Chatbots

(15) What is Large Language Model (LLM)

→ is an AI model trained on large amounts of text so it can learn patterns in language & generate or transform text.

→ A standard machine learning model may be designed for a specific task.

→ While LLM can perform many language-related tasks.

→ Two examples are ChatGPT & Claude, which can answer questions, summarise info, generate text & assist with coding.

(16) What is Generative AI?

- An AI that can create new content based on patterns learned from training data.
- At high level, the system learns patterns from large amount of existing information & generates new output based on a user's request.
- Examples include ChatGPT for text, DALL-E for images & GitHub Copilot for coding assistance.

(17) What is training Dataset?

- is a collection of information used to teach machine learning or AI model.
- Its quality is important because the model learns patterns from it.
- If the data is incorrect, incomplete or biased, the model may produce incorrect or unfair results.
- E.g. A recruitment AI trained on biased historical applications can learn & repeat that bias.

(18) What is computer vision?

- A field of AI that allows computers to analyse & interpret images & video.
- Machines process images as data & use patterns learned from training data to identify objects, faces, text or other features.
- E.g. - Facial recognition for identity verification & vehicle cameras recognising roads, vehicles & pedestrians.

Section C

(19) What is an API?

- API stands for Application Programming Interface.
- Allows different software to exchange information.
- Simple analogy is a restaurant: the customer is like the application making request, the waiter is like the API, and the kitchen is like the other system.
- The waiter takes the request to the kitchen & brings the result back.
- APIs are important because they allow systems to work together.

(20) What is an API key?

- is a unique piece of information used to identify or authenticate an application when it accesses an API.
- It should be treated like a password & stored securely.
- It should not be posted publicly on websites or GitHub repositories.
- If exposed, another person could potentially use the service associated with it, causing unauthorized access, costs or data exposure.

(21) What is a REST API?

- REST = Representational State Transfer
- A REST API is a common way for applications to communicate over the internet using HTTP.
- GET is used to retrieve information.
- POST to send or create information.
- PUT to update or replace information, & .
- DELETE to remove information.

(22) What is JSON?

- It is a lightweight format used to store & exchange data.
- It is popular with APIs because it is easy for computers to process & relatively easy for people to read. e.g.

```
{ "name": "Katie", "age": 32, "occupation": "..." }
```

(23) What is prompt Engineering?

- Is a process of writing clear & effective instructions for AI system so it produces a useful response.
- It matters because clarity can affect the quality of results.
- Poor prompt: "Explain SQL"
- Improved prompt "Explain SQL to a beginner who has not worked with databases".
- Poor prompt: "Make me a CV"
- Improved prompt would specify the target role, experience & ATS-friendly format.

(24) What is a MAP function?

- is used in programming & data processing to apply an operation to each item in a collection.
- Example, mapping a list $[1, 2, 3, 4]$ by multiplying every number by 2 produces $[2, 4, 6, 8]$.
- Map means taking each item & applying the same instruction to it.

(25) What is cloud Computing?

- using computing resources such as storage, servers & software through the internet instead of owning & maintaining all the physical infrastructure.
- **IaaS** (Infrastructure as a Service)
 - provides infrastructure such as virtual servers & storage;
 - e.g. Amazon EC2.
- **PaaS** (Platform as a Service)
 - provides an environment for building & running applications.
 - e.g. Google App Engine.

(1) SaaS (Software as a Service)

- provider complete software through the internet
- e.g. Microsoft 365

(26) What is Version Control System

- Records changes made to files over time, allowing users to see previous versions & return to them if necessary.
- Git is a popular version control system used by developers & data professionals to track work & collaboration.
- A commit is like saving a snapshot of changes.
- A branch is a GitHub host Git repositories online & support collaboration.
- It can store SQL scripts, notebooks, datasets, projects & documents.

(27) What is Jupyter Notebook?

- is an interactive environment where users can write & run programming code, view results & add explanations in the same document.
- It is mostly associated with Python in data analytics.
- It is popular because analysts can combine code, data, charts & written explanations in one place.

(28) What is Python?

- is a general-purpose programming language that is popular because it is relatively easy to learn & has many libraries for data analysis, automation, machine learning & AI.
- Pandas is used for tabular data, NumPy for numerical calculations & arrays, Matplotlib for charts & visualisations, & Scikit-learn for machine learning.

Section 1

(29) What is Data privacy?

- It's about protecting people's personal information & making sure it is collected, stored & used responsibly.
- It is important because personal information can be sensitive & poor handling can lead to identity theft, financial loss or loss of trust.
- The GDPR, or General Data Protection Regulation, is a European Union data protection law.
- It can also apply to organisations outside EU when they offer goods or services to people in the EU or monitor their behaviour.

(30) What is Data Ethics?

- Means using data & AI in a responsible, fair & respectful way.
- Examples of unethical use:
 - (a) Selling personal information without permission
 - (b) Using biased data to make decisions about people,
 - (c) Using AI to create or spread false information
- These are problematic because they can violate privacy, cause unfair treatment or mislead people.

(31) What is Data Governance?

- Is a system of rules, processes & responsibilities used to manage an organisation's data.
- Good governance includes data quality, security, privacy, access controls, ownership & responsibility, policies & standards, & compliance with laws.
- The goal is to make sure data is accurate, secure, available to the right people & used appropriately.

(32) What is Bias in AI?

- Bias occurs when an AI system produces unfair or inaccurate results because of problems in its data, design or assumptions.
- Example, a recruitment system trained mostly on historical applications from one group of successful employees could learn that pattern & unfairly rank similar applicants higher.
- Consequences can include, discrimination, lost opportunity & poor org. organisational decision.

(33) Data Analyst vs Data Scientist vs Data Engineer vs AI Engineer.

(a) Data Analyst

- Analyses data & creates insight & reports; common skills
- * Common Skills include Excel, SQL, Power BI & Python.

(b) Data Scientist

- Builds advanced models & make predictions.
- * Common Skills include Python, Statistics & Machine Learning.

(c) Data Engineer

- Builds & maintain systems that collect & process data
- * Common Skills include SQL, Python, cloud platforms & databases.

(d) AI Engineer

- Builds & implements AI systems & applications
- * Common Skills include Python, Machine Learning, AI Tools & APIs

(34) What is Business Intelligence? (BI)

- It is the process of using data, reports & visualisation to help org. organisations make better business decisions.
- Two popular BI tools are Microsoft Power BI & Tableau.
- E.g. a company can use Power BI to create a dashboard showing monthly sales, expenses & profit. Managers
- Managers can use the dashboard to identify problems & make decisions based on data.

Section 2.

(35) Interesting or Surprising Concept

* Machine Learning (ML)

→ Because it is fascinating that computers can learn patterns from data instead of being programmed with every rule.

(36) Most difficult concept

* JSON

→ I found it difficult to understand how information is stored using key-value pairs & how the data is structured.

(37) How do you see Data & AI affecting your future career or everyday life?

→ Data & AI will have a huge positive impact in my life because I am in finance industry, using spreadsheets & databases while I am trying to move towards Data Analytics.

→ I already work with financial & operational information, so learning how to analyse data could help me identify trends, errors, unnecessary costs & opportunities for improvement.

→ I can use tools like SQL, Excel, Power BI & eventually Python to make work more efficient.